The Environment in Four Equations

Simple reference sheet (not part of the paper)

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The whole simulated world is specified by four choices: how methane spreads, how the wind behaves, what the sensors record, and what the models are told. Each is one equation. Nothing else exists in the environment.

1. How methane spreads from the source (transport)

A leak of rate q (kg/h) at location c produces at each sensor and time step a concentration given by the steady-state Gaussian plume:

$$g \;=\; \frac{1}{2\pi\, U\, \sigma_y \sigma_z}\, \exp\!\left(-\frac{y_{\mathrm{cw}}^2}{2\sigma_y^2}\right) \left[\exp\!\left(-\frac{(z_r - H)^2}{2\sigma_z^2}\right) + \exp\!\left(-\frac{(z_r + H)^2}{2\sigma_z^2}\right)\right] \tag{1}$$

In words: the wind carries the gas downwind at speed U; turbulence spreads it sideways and vertically into a widening cone (the widths σ_y, σ_z grow with downwind distance, faster in unstable air, slower in calm stable air); the ground reflects it back up (the second exponential); and nothing exists upwind of the leak ($g = 0$ there). Every property of the site's atmosphere and geometry lives in this one formula.

2. How the wind behaves (the driver)

The wind is not constant. Each scenario draws a mean speed and direction, and the wind wobbles around them from step to step as a first-order autoregressive (AR(1)) meander:

$$\theta_t = \theta_0 + A_{\mathrm{dir}}\, e_t, \qquad U_t = U_0\, e^{A_{\mathrm{spd}}\, e'_t}, \qquad e_t = \rho\, e_{t-1} + \sqrt{1-\rho^2}\; \varepsilon_t \tag{2}$$

with correlation $\rho = 0.7$, direction wobble A_{dir} of 20–40°, and speed wobble A_{spd} of 10–30%.

In words: the plume swings back and forth across the site over the 30 time steps. This meander is the one dynamic element of the environment, and it is what lets a handful of fixed masts see the plume from many angles.

3. What the sensors record (observation)

Readings are linear in the leak rate, plus noise:

$$y \;=\; q\, g_{c^*} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I) \tag{3}$$

In words: each mast records the true plume concentration at its location, corrupted by Gaussian sensor noise (scale σ from 0.2 to 3.0 ppm, spanning low-cost to mid-grade sensors), and up to 20% of readings are randomly missing. Doubling the leak doubles every reading; that linearity is what makes the exact posterior computable.

4. What the models are told (information)

The dataset records the anemometer's *measurement* of the wind, never the truth. At every step the measured direction and speed are corrupted:

$$\hat{\theta}_t = \theta_t + \delta_t, \quad \delta_t \sim \mathcal{N}(0, 10^{\circ\,2}), \qquad \hat{U}_t = U_t\, e^{\eta_t}, \quad \eta_t \sim \mathcal{N}(0, 0.10^2) \tag{4}$$

In words: a real site only has a wind sensor, and inexpensive anemometers are good to about 10° in direction and 10% in speed. No model, at any stage, observes the true wind.

Why this ordering tells the story

Items 1–3 define the physical world; item 4 defines the information gap. The method's entire job is to close item 4's gap using item 1's formula: the physics input images are Eq. (1) evaluated under many plausible winds drawn from Eq. (4)'s error model, summarized into evidence, visibility, fragility, and fit quality at every candidate cell.